

ZALAU, Romania

WMO: 150630

Lat: 47.1950N

Lon: 23.0467E

Elev: 296

StdP: 97.82

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.9	-15.3	1.0	-8.5	-12.7	1.3	-5.9	8.7	0.5	7.5	0.6	1.1	140	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.4	31.8	20.3	30.2	20.0	28.6	19.4	21.6	28.9	20.9	28.1	20.2	26.8	1.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	14.7	24.1	18.6	13.9	23.3	17.9	13.3	22.9	64.2	29.0	61.5	28.2	59.1	27.0	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.1	5.0	-13.4	33.5	2.5	2.3	-15.2	35.2	-16.7	36.5	-18.1	37.8	-19.9	39.4		
(5)			-13.9	23.3	2.3	0.9	-15.5	24.0	-16.9	24.5	-18.2	25.0	-19.8	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	-0.4	2.4	5.6	11.3	15.9	19.5	20.9	21.5	16.5	11.3	6.1	0.8	(6)
(7)		DBStd	8.66	4.29	4.48	4.38	4.64	3.61	3.36	2.98	3.44	4.46	4.29	4.69	4.29	(7)
(8)		HDD10.0	1186	323	214	146	40	3	0	0	0	3	38	134	286	(8)
(9)		HDD18.3	2989	581	446	394	212	92	26	12	9	86	219	367	544	(9)
(10)		CDD10.0	1551	0	1	9	80	185	286	339	356	199	78	17	1	(10)
(11)		CDD18.3	311	0	0	0	2	16	62	92	106	32	1	0	0	(11)
(12)		CDH23.3	2727	0	0	0	18	124	502	806	1019	250	8	0	0	(12)
(13)		CDH26.7	852	0	0	0	1	12	130	247	395	67	0	0	0	(13)
(14)	Wind	WSAvg	1.6	1.5	1.7	1.9	2.0	1.7	1.6	1.5	1.5	1.5	1.4	1.7	1.5	(14)
(15)	Precipitation	PrecAvg	654	34	30	37	48	72	95	86	56	63	49	45	45	(15)
(16)		PrecMax	958	58	77	118	85	155	189	185	102	138	100	94	90	(16)
(17)		PrecMin	389	9	9	8	8	22	16	19	6	7	4	0	7	(17)
(18)		PrecStd	139	14	16	25	19	37	43	44	26	32	28	21	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	14.9	19.5	25.5	28.0	31.6	32.5	34.6	31.6	24.5	18.4	11.9	(19)
(20)			MCWB	7.6	8.9	11.2	15.0	17.3	21.6	20.5	20.5	20.0	15.7	12.6	8.1	(20)
(21)		2%	DB	8.4	12.2	16.4	23.0	26.2	29.9	31.1	32.8	28.3	22.7	16.5	9.4	(21)
(22)			MCWB	5.7	7.8	9.2	13.7	17.0	20.5	20.4	20.1	18.4	14.9	11.5	6.9	(22)
(23)		5%	DB	6.7	10.6	14.1	20.8	24.6	28.0	29.4	30.8	26.1	20.4	14.8	7.9	(23)
(24)			MCWB	4.6	6.9	8.1	12.6	16.2	19.5	19.9	19.7	17.4	14.2	10.7	5.8	(24)
(25)		10%	DB	5.1	9.0	12.1	18.8	22.8	26.3	27.7	28.7	23.9	17.9	12.9	6.2	(25)
(26)			MCWB	3.3	5.9	7.3	11.5	15.3	19.1	19.3	19.4	16.9	13.2	9.7	4.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	9.8	12.0	15.9	19.1	23.2	22.3	22.2	21.0	17.2	13.2	8.9	(27)
(28)			MCDWB	10.2	14.2	18.6	23.9	25.7	29.4	29.3	30.7	30.0	22.1	17.3	11.0	(28)
(29)		2%	WB	6.0	8.3	10.0	14.5	17.8	21.5	21.5	21.3	19.3	15.8	12.0	7.2	(29)
(30)			MCDWB	7.9	11.5	14.9	21.7	24.3	27.4	28.5	29.8	26.1	20.3	15.4	9.3	(30)
(31)		5%	WB	4.7	7.3	8.8	13.3	16.9	20.5	20.7	20.6	18.3	14.8	11.0	5.8	(31)
(32)			MCDWB	6.5	10.1	13.1	19.4	23.0	26.6	27.5	28.5	24.5	19.5	14.4	7.6	(32)
(33)		10%	WB	3.5	6.3	7.7	12.3	16.0	19.6	20.0	19.9	17.3	13.7	10.0	4.9	(33)
(34)			MCDWB	4.9	8.8	11.7	17.6	21.7	25.2	26.1	27.1	22.8	17.3	12.9	6.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	6.5	8.3	9.8	9.8	10.0	10.7	11.4	9.6	8.8	6.2	4.7	(35)
(36)			MCDBR	6.5	8.7	11.0	12.5	12.5	12.3	13.3	13.6	12.7	12.5	8.5	6.2	(36)
(37)			MCWBR	4.4	5.3	5.9	5.7	5.3	4.9	4.8	4.3	4.8	5.9	4.8	4.5	(37)
(38)		5% WB	MCDWB	6.1	8.1	9.7	11.6	11.2	11.2	11.8	12.3	11.4	10.7	7.8	5.9	(38)
(39)			MCWBR	4.3	5.2	5.7	5.6	5.2	4.9	4.6	4.4	4.7	5.5	4.5	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.319	0.346	0.386	0.434	0.420	0.418	0.437	0.438	0.401	0.371	0.344	0.319	(40)	
(41)		taud	2.324	2.278	2.221	2.120	2.225	2.261	2.219	2.213	2.282	2.343	2.364	2.358	(41)	
(42)		Ebn at Noon	751	806	826	818	846	848	825	806	802	768	714	705	(42)	
(43)		Edh at Noon	75	97	119	145	136	132	136	131	112	90	73	65	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.09	1.91	3.12	4.43	5.40	5.91	5.80	5.33	3.73	2.45	1.36	0.83	(44)	
(45)		RadStd	0.16	0.28	0.46	0.46	0.54	0.42	0.51	0.48	0.44	0.30	0.19	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		+0.69	N/A	N/A	+1.03	N/A	N/A	-134	-198	+112	+55

Nomenclature: See separate page